



BITS Pilani
Pilani Campus



SURVEYING (CE F213)

(Vertical Distance Measurement : Levelling)

Objectives to learn

Different terms related to levelling , Purpose of levelling, Concept and principles, types of levels and staffs, field procedure, checks etc.

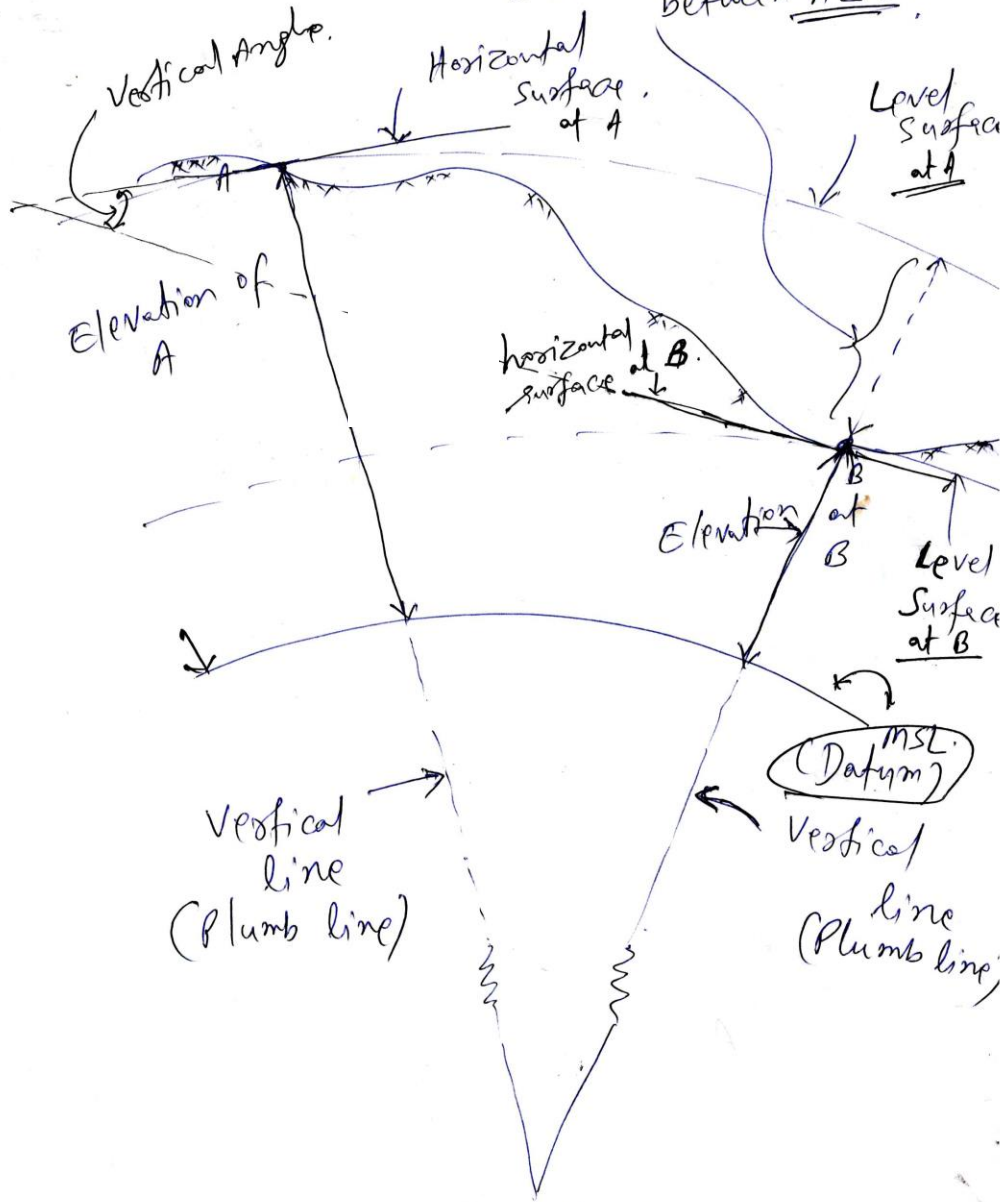
Levelling.

Difference of elevation
between A & B.

innovate

achieve

lead



When curvature of
earth is neglected (vertical lines are parallel)

• Centre of earth.

Definition of Levelling: Levelling is defined as “an act of determining the relative height of different points on, above or below the earth surface”.

Different terms used in vertical distance measurement and their definitions :

1. **Level surface:** Any surface parallel to mean spheroidal surface of the earth is called as level surface. The mean spheroidal surface of the earth is also known as geoid. Example: the water surface in a still lake. It is curved surface. Every point on this surface is of equal distance from the center of the earth. The normal to the surface at any point is plumb line.
2. **Level line:** Any line on the level surface is called the level line. It is a curved line. At any point on the level line the tangent to the level line and the plumb line(the direction of gravity) are perpendicular to each other.

3. **Vertical line:** Any line from any point on the earth surface to the center of the earth is called as vertical line. It is indicated by the plumb line(the direction of gravity).
4. **Horizontal Plane:** It is a tangential plane at any point on the level surface. This plane is perpendicular to the plumb line.
5. **Horizontal line:** Any line on the horizontal plane is a horizontal line.
6. **Datum:** It is a reference surface having zero elevation, wrt this elevations of other points are calculated.
7. **Elevation(Reduced Level):** It is the vertical distance of a point above or below the datum.
8. **Height of Instrument:** It is the elevation of the plane of collimation when the instrument is leveled.

Levelling

9. **Back Sight(BS):** It is the staff reading at a point with known elevation(called bench mark)
10. **Fore Sight(FS):** It is the last staff reading at a point if the instrument is not shifted or before shifting the instrument.
11. **Intermediate sight:** Any staff reading between the back sight and fore sight is called Intermediate sight(IS).
12. **Change Point(CP):** It is a point which denotes the change of instrument. It is also called **Turning Point(TP)**.
13. **Bench Mark:** It is a point where the elevation is known.
14. **Station:** It is any point whose elevation is calculated or to be calculated.

MSL : Mean Sea Level

Types of Levels:

1. Dumpy level
2. Wye level
3. Tilting level
4. Digital level
5. Automatic Level
6. Hand level

Levelling

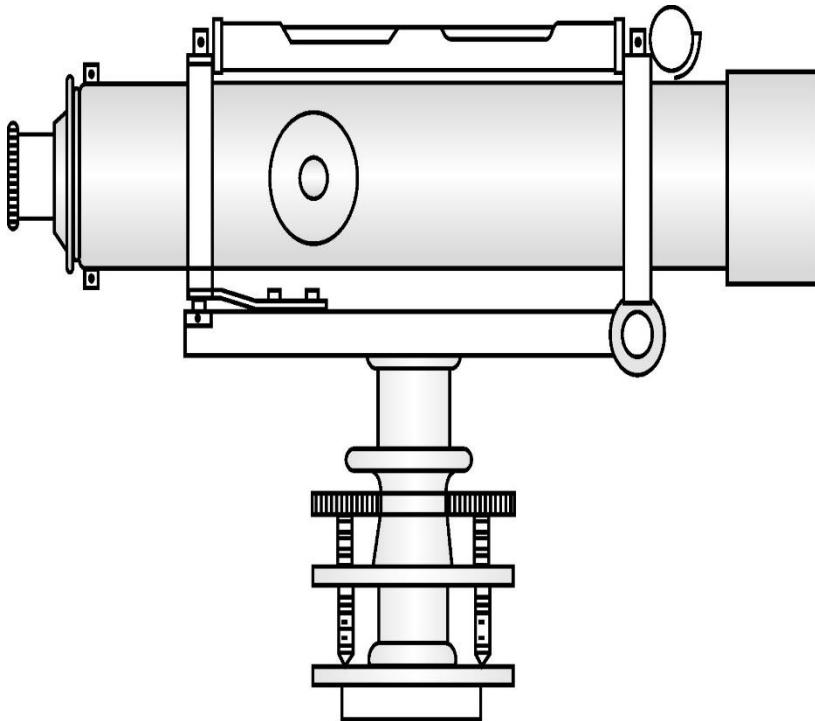
Dumpy level



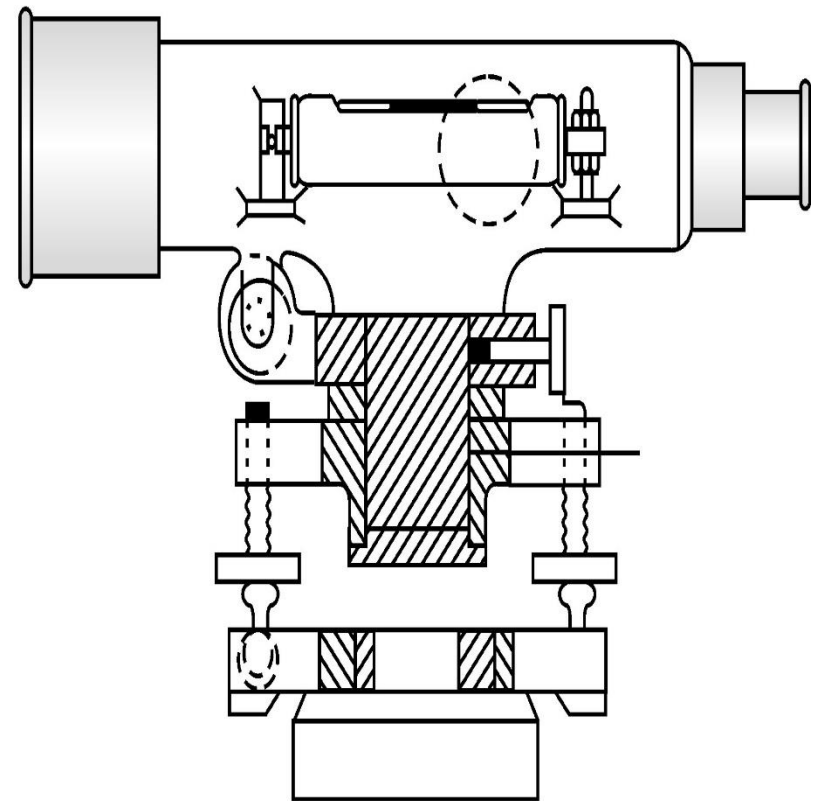
Automatic level



Levelling



(a) Dumpy level



(b) Sectional view

levelling



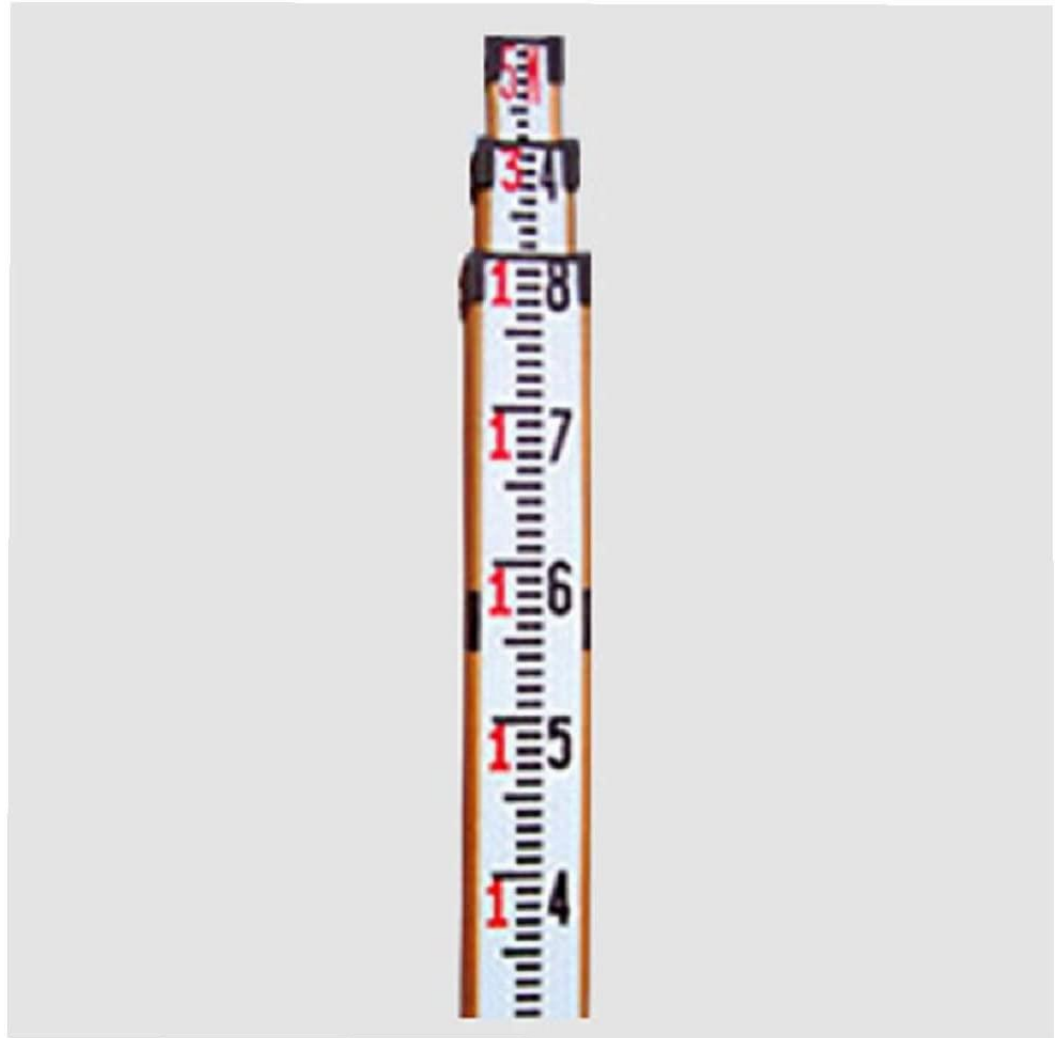
Types Levelling Staff

- 1.
- 2.
- 3.
- 4.

levelling

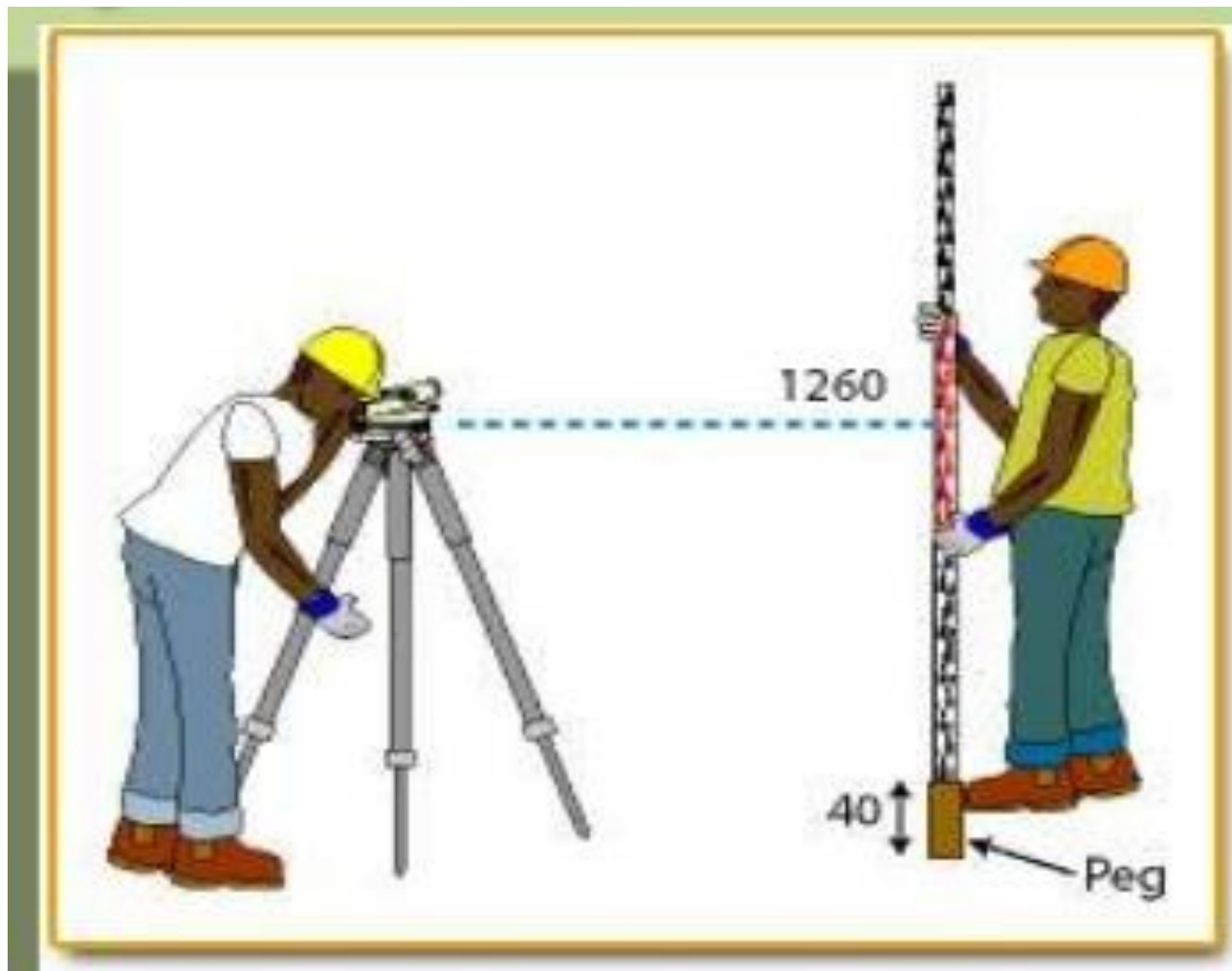


Least count is 0.005m(5mm)



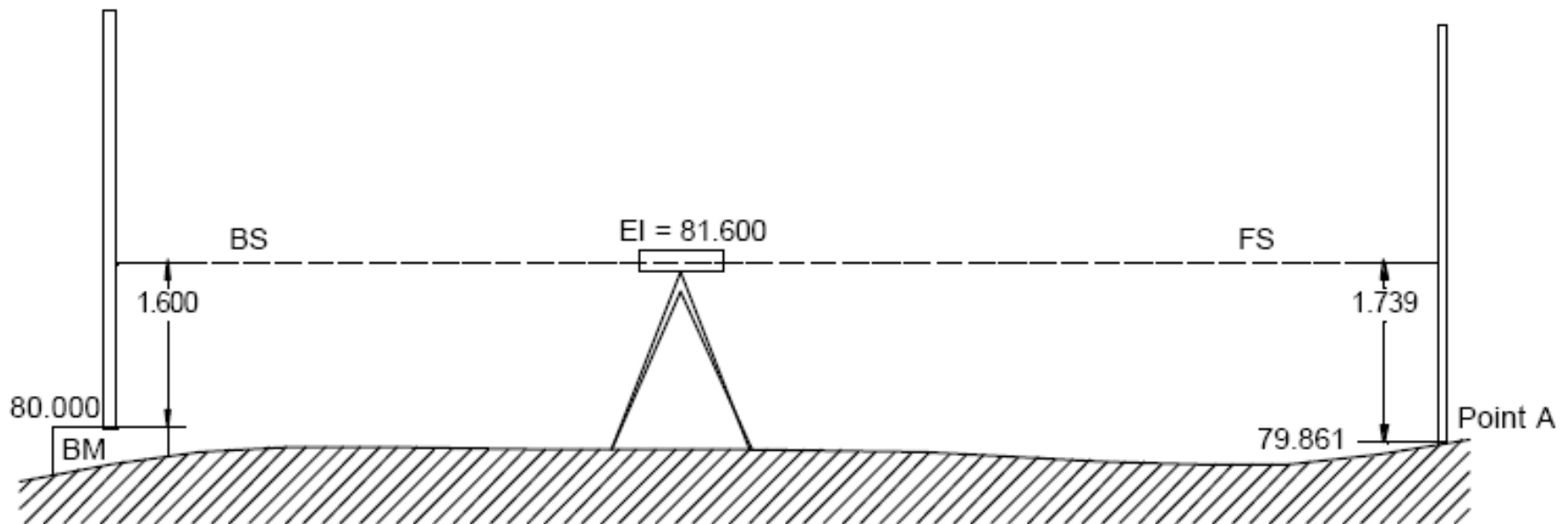
Types of levelling

1. Simple levelling
2. Differential levelling
3. Fly levelling
4. Check levelling
5. Profile levelling
6. Cross levelling
7. Reciprocal levelling
8. Precise levelling
9. Trigonometric levelling
10. Barometric levelling
11. Hypsometric levelling

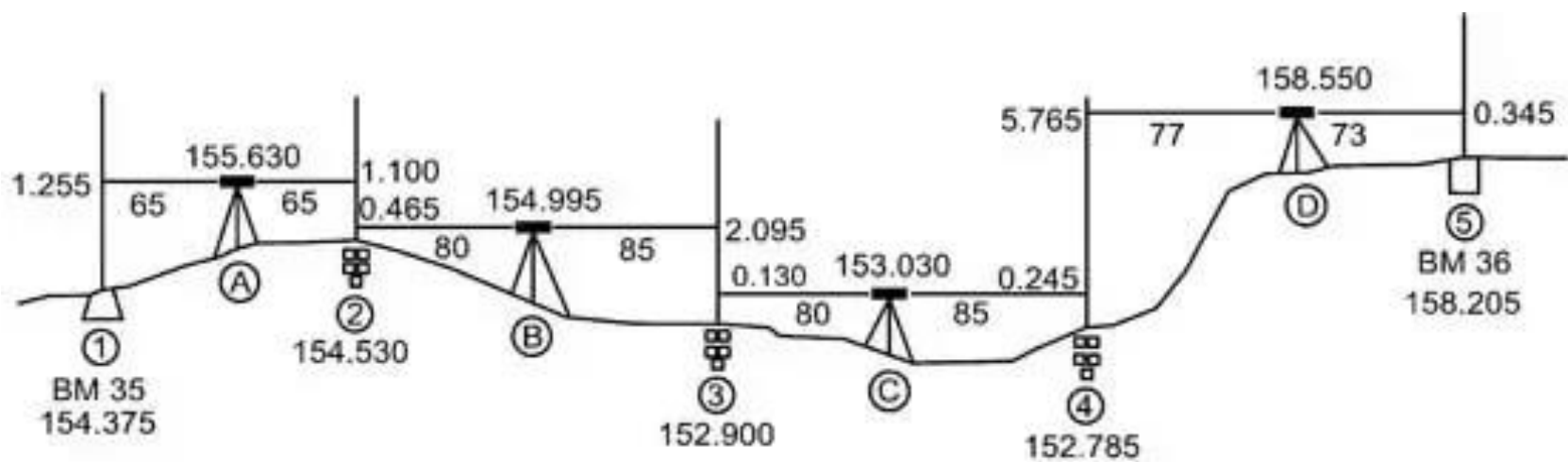




Simple levelling:-It is the simplest method used, when it is required to find the difference in elevation between 2 points.



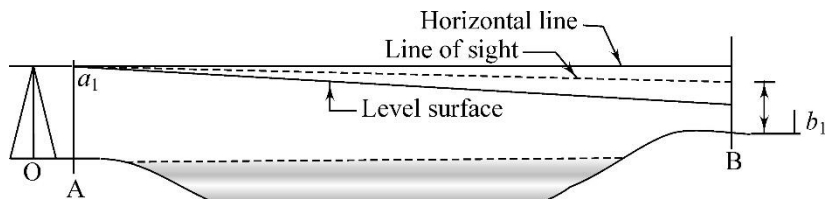
Differential Levelling: This method is used to find the difference in the elevation between points if they are too far apart or the difference in elevation between them is too much or points having obstacle in between.



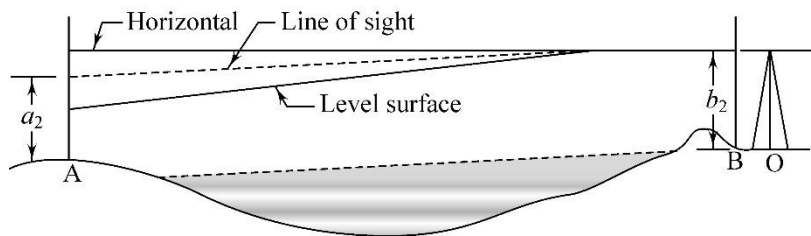
Levelling



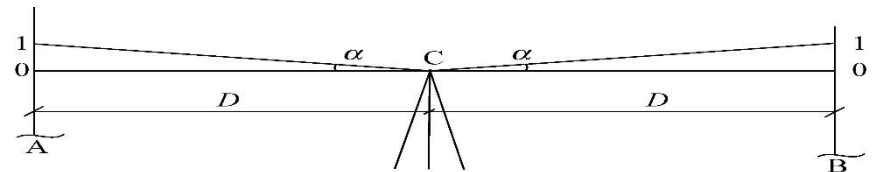
RECIPROCAL LEVELLING



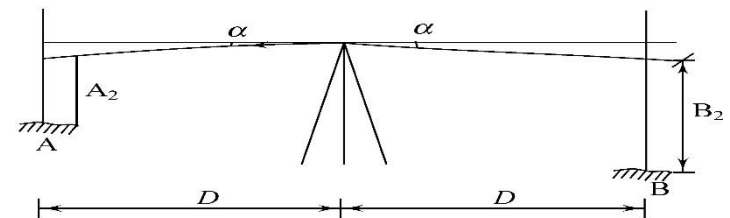
(a) Instrument at A



(b) Reciprocal levelling



(a) Line of sight inclined upwards



(b) Line of sight inclined downwards

Will be discussed in detail in the following classes.

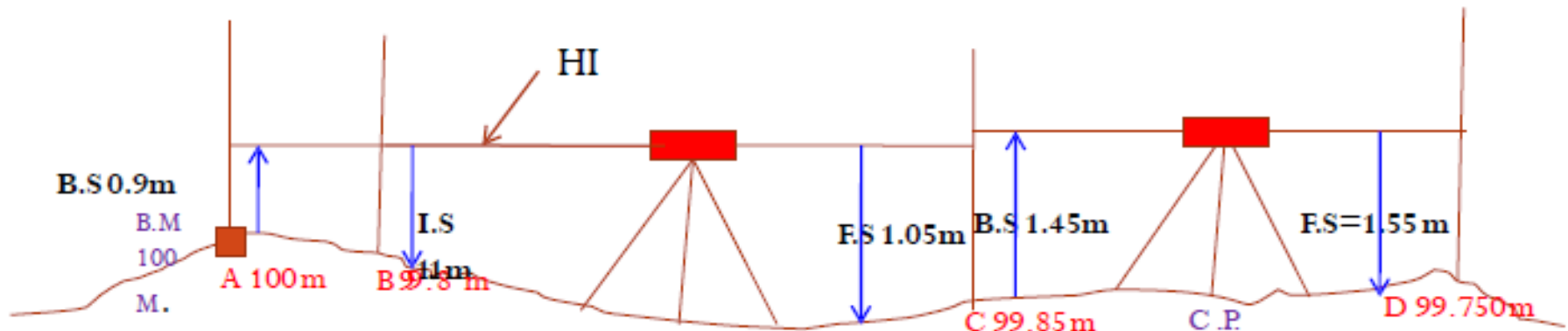
Method to calculate RL(Elevation)

1. **Height of Instrument Method(Collimation method):-** This method consist of finding H.I. for every setup of instrument, and then obtaining the R.L. of point of reference with respect to H.I.
2. **Rise and fall method:-** This method consists of determining the difference of elevation between consecutive points by comparing each point with immediate preceding point.

Levelling



Height of Instrument Method(Collimation method)



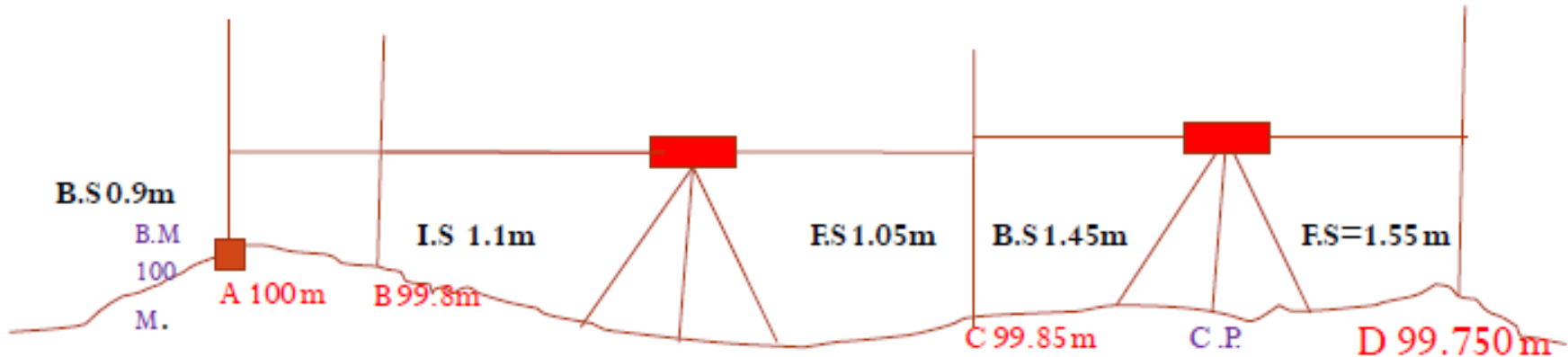
Station	B.S	I.S	F.S	H.I	R.L	Remark
A	0.9			100.9	100.00	B.M
B		1.1			99.800	
C	1.450		1.05	101.3	99.850	C.P.
D			1.550		99.750	

$$\Sigma BS - \Sigma FS = last\ RL - first\ RL$$

Levelling



Rise and fall method



Station	B.S	I.S	F.S	Rise	Fall	R.L	Remark
A	0.9					100.00	B.M
B		1.1			0.2	99.800	
C	1.450		1.05	0.05		99.850	C.P.
D			1.550		0.1	99.750	

$$\Sigma BS - \Sigma FS = \text{last RL} - \text{first RL} = \Sigma \text{rise} - \Sigma \text{fall}$$

Levelling

Q1: The following readings were taken in sequence during a levelling work. The first reading was taken on a bench mark of assumed elevation 100.000. Find the R.L.s of the remaining stations if the instrument was shifted after the **third, seventh, tenth and thirteenth** readings. Use the height of collimation method and rise and fall method.

1.605, 2.150, 1.385, 1.895, 1.365, 2.105, 1.950, 0.985, 1.305, 1.185, 1.305, **-2.105**, 1.385, 1.005, 1.155 and 1.145.

Levelling

Height of Instrument Method(Collimation method)

Station	B.S.	I.S.	F.S.	Height of Instrument	Reduced level	Remarks
1	1.605			101.605	100.000	B.M.
2		2.150			99.455	
3	1.895		1.385	102.115	100.220	
4		1.365			100.750	
5		2.105			100.010	
6	0.985		1.950	101.15	100.165	
7		1.305			99.845	
8	1.305		1.185	101.27	99.965	
9		-2.105			103.375	
10	1.005		1.385	100.89	99.885	
11		1.155			99.735	
12			1.145		99.745	

$$\Sigma BS - \Sigma FS = last\ RL - first\ RL$$

Rise and fall method

Q2: In a levelling operation, the following observations are obtained keeping the dumpy level at four different positions. In the 1st position of the dumpy level, the 1st reading is taken on a benchmark. In the 4th position of the level, the last reading is taken on the same benchmark. Find the R.L.s of the stations and the last staff reading using Rise and Fall method. Perform all checks.

Dumpy level position	Consecutive reading on staff(m) at different stations	Remark
1	2.245, 1.460, 2.550, 2.070, 1.655 and 1.765	The 1 st reading is taken on a benchmark with R.L. 200.120m
2	3.440, 1.980, 2.160, 1.885, 1.965 and 1.775	
3	2.890, 1.340, 2.555, 2.995 and 1.835	
4	2.115, 2.005, 1.975, 1.645 and the last reading	The last reading is taken on the same benchmark.

No. 2.

(Staff) Station	B.S. (m)	I.S. (m)	F.S. (m)	Rise (m)	Fall (m)	R.L. (m)	Remarks
A	2.245					200.12	B.M.
B		1.460		0.785		200.905	
C		2.550			1.09	199.815	
D		2.070		0.48		200.295	
E		1.655		0.415		200.71	
F	3.440		1.765		0.11	200.6	C.P.
G		1.980		1.46		202.06	
H		2.160			0.18	201.88	
I		1.885		0.275		202.155	
J		1.965			0.08	202.075	
K	2.890		1.775	0.19		202.265	C.P.
L		1.340		1.55		203.815	
M		2.555			1.215	202.6	
N		2.995			0.44	202.16	
O	2.115		1.835	1.16		203.32	C.P.
P		2.005		0.11		203.43	
Q		1.975		0.03		203.46	
R		1.645		0.33		203.79	
S			5.315		3.67	200.12	B.M.
$\Sigma F.S.$ = 10.69			$\Sigma F.S.$ = 10.69	$\Sigma Rise$ = 6.785	$\Sigma Fall$ = 6.785		

(17) with proper calculation

Normally the staff height is 3 to 5m. So if we keep the staff at the B.M. point it will need a reading of this. (17 + 2 + 1) 17 ← 17 R.L. 2 ← last reading 1 ← checks.



Q3: The following consecutive readings are taken in m with a dumpy level and a 4m levelling staff.

1.455, 2.565, 1.995, 2.005, 1.375, 1.885, 3.125, 2.050, 1.675, 2.135, 1.395, 3.205, 2.275, 1.635, 1.935, 1.435, 1.535, 2.245, 1.765, 1.555, 1.745, 2.335, 1.455 and 1.615.

The instrument is shifted after 4th, 8th, 13th, 17th and 22nd readings. The R.L. of the staff station where the 15th reading taken is 305.93m. Find the R.L. of all staff stations using rise and fall method. Perform all checks.

No. 2.

Staff Station	B.S. (m)	I.S. (m)	F.S. (m)	Rise (m)	Fall (m)	R.L. (m)	Remarks
1	1.455			1.11		308.055	
2		2.565			0.57	306.945	
3		1.995		0.01		307.515	
4	1.375		2.005	0.51		307.505	C.P
5		1.885		1.24		306.995	
6		3.125			1.075	305.755	
7	1.675		2.050	0.46		306.83	C.P
8		2.135			0.74	306.37	
9		1.395		1.81		307.11	
10		3.205			0.93	305.3	
11	1.635		2.275	0.3		305.93 306.23	C.P
12		2.935				305.93	B.M.
13		1.435		0.5		306.43	
14	2.245		1.535		0.1	306.33	C.P
15		1.765		0.48		306.81	
16		1.555		0.21		307.02	
17		1.745			0.19	306.83	
18	1.455		2.335		0.59	306.24	C.P
19			1.615		0.16	306.08	
$\Sigma B.S. = 9.84$				$\Sigma F.S. = 11.815$	$\Sigma Rise = 6.63$	$\Sigma Fall = 4.355$	

innovate

achieve

lead

Checks.

(i) $\sum B.S. - \sum F.S. = 9.84 - 11.815 = -1.975 \text{ m}$

(ii) $\text{Last R.L.} - 1^{\text{st}} \text{ R.L.} = 306.08 - 308.055 = -1.975$

(iii) $\sum \text{Rise} - \sum \text{Fall} =$ To find this, we need to

(3) calculate the rises and falls considering the 1st observation as the reference point. In this case the rises and falls are calculated w.r.t to intermediate point where the B.M. is given. Calculation of R.L.s are ok.

Effect of Curvature of Earth and Refraction

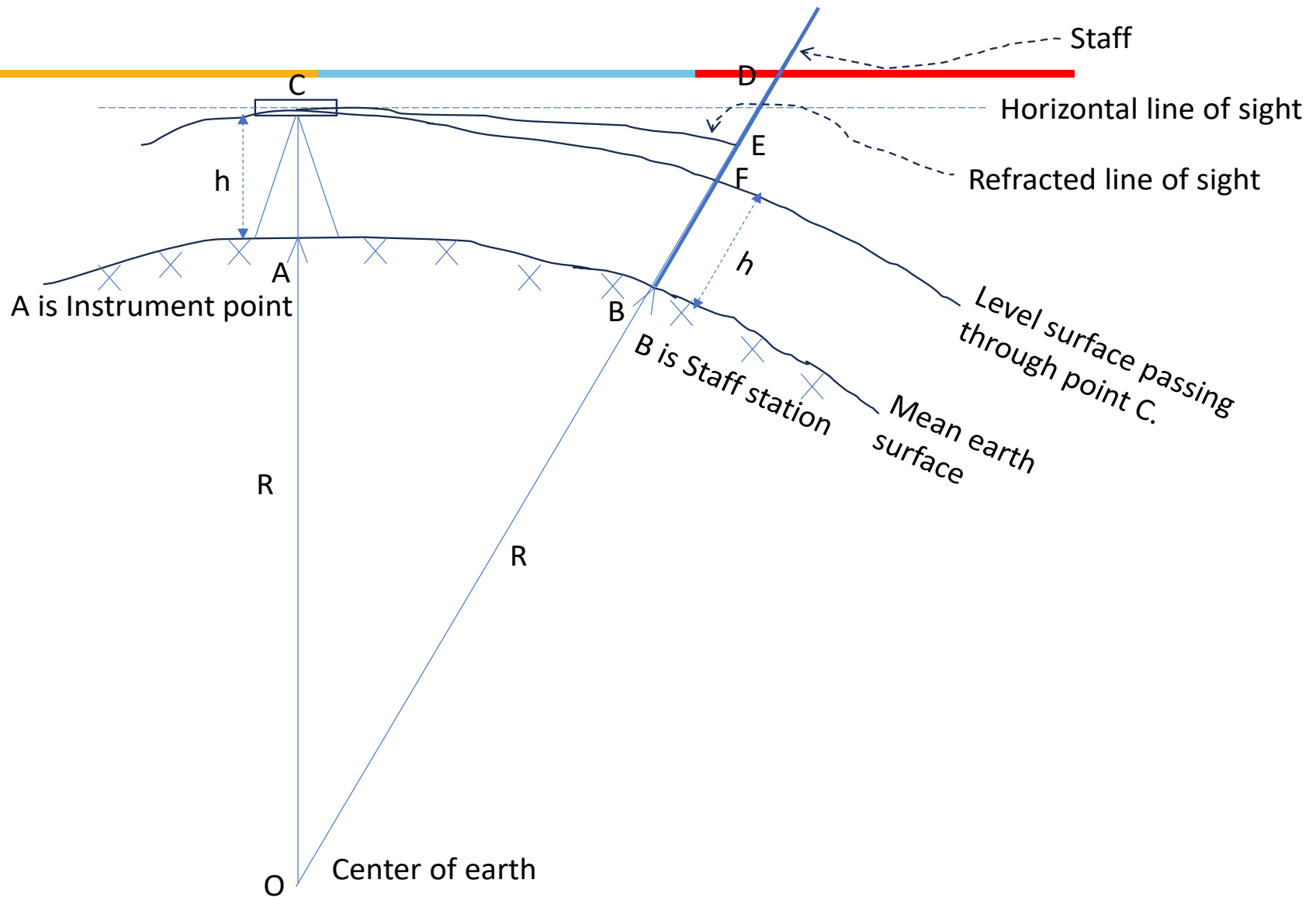
These effects are to be considered :

- (i) In precise levelling
- (ii) If the stations are far away from one another or the instrument is far away for the staff stations.

The effect of curvature of earth

The curvature increase the staff reading, so the RL of the station appears lower. So the error is positive(+ve).

Correction: The staff reading should be corrected. The correction is subtractive(-ve).



$$OD^2 = OC^2 + CD^2$$

$$(OB + BF + FD)^2 = (OA + AC)^2 + CD^2$$

$$((R + h) + FD)^2 = (R + h)^2 + CD^2$$

$$(R + h)^2 + FD^2 + 2(R + h)FD = (R + h)^2 + CD^2$$

$$FD^2 + 2(R + h)FD = CD^2$$

$$FD(FD + 2(R + h)) = CD^2$$

$$FD = \frac{CD^2}{(FD + 2(R + h))} \quad h \ll R \quad FD \ll R$$

$$FD = \frac{CD^2}{2R} \quad R = 6370 \text{ km}$$

$CD = D = \text{distance from the instrument point to staff station in km}$

$FD = \text{error due to curvature} = C_c$

$$FD = C_c = 7.85 \times 10^{-5} D^2 km$$

$$C_c = 0.0785 D^2 m \quad D \text{ is in } km$$

The effect of Refraction

Refraction decreases the staff reading, so the RL of the station appears higher.
The error is negative(-ve).

Correction : The correction on the staff reading is additive(+ve)

$$\begin{aligned}\text{Error due to refraction} &= \frac{1}{7} \text{ of error due to curvature} \\ &= \frac{1}{7} \times 0.0875 D^2 \text{ m} \\ &= 0.0112 D^2 \text{ m}\end{aligned}$$

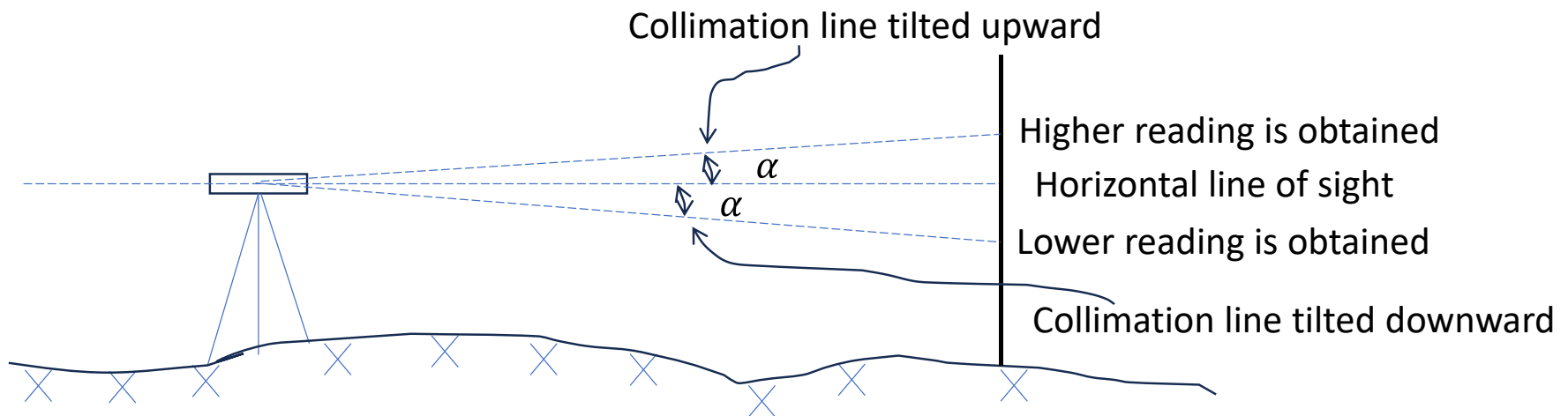
D is in km

True staff reading = Observed staff reading – curvature correction + refraction correction

$$\begin{aligned}\text{True staff reading} &= \text{Observed staff reading} - 0.0875 D^2 + 0.0112 D^2 \\ &= \text{Observed staff reading} - 0.0673 D^2\end{aligned}$$

where **D** is in km

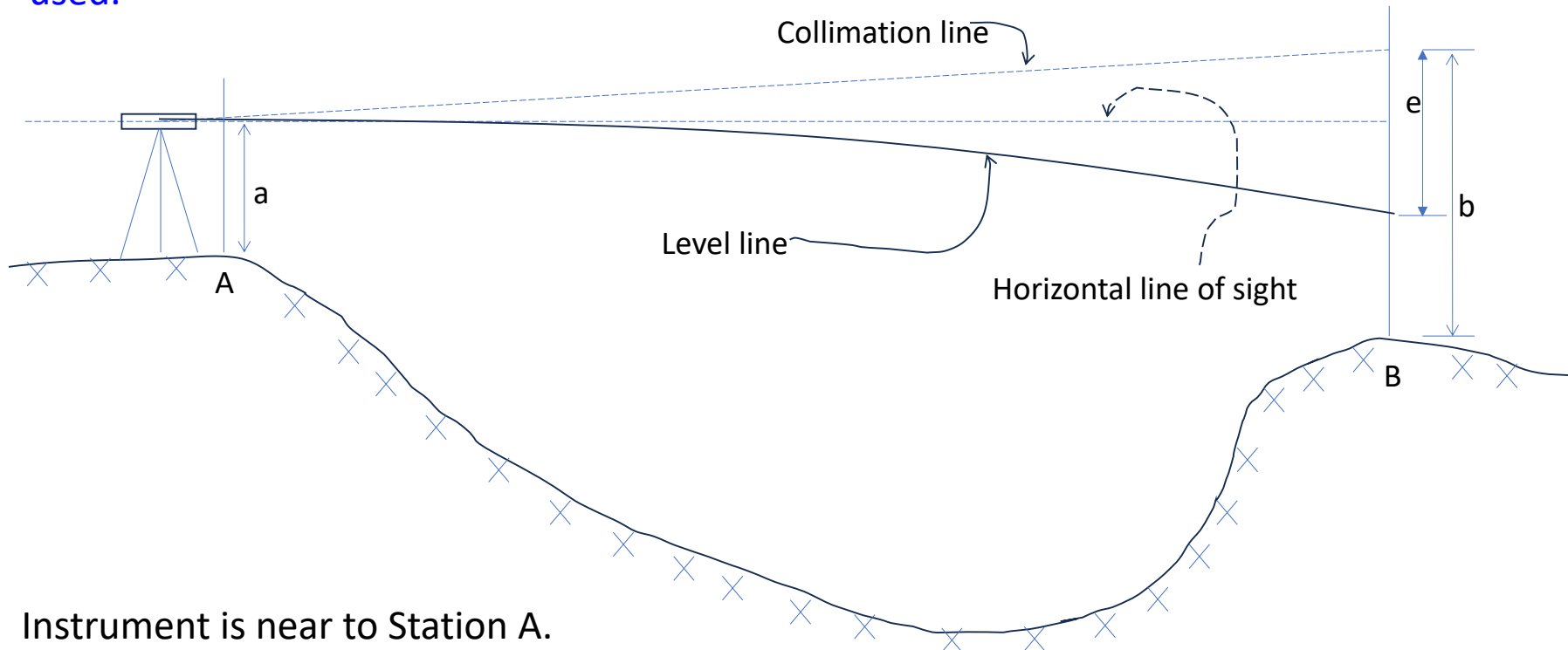
Collimation error



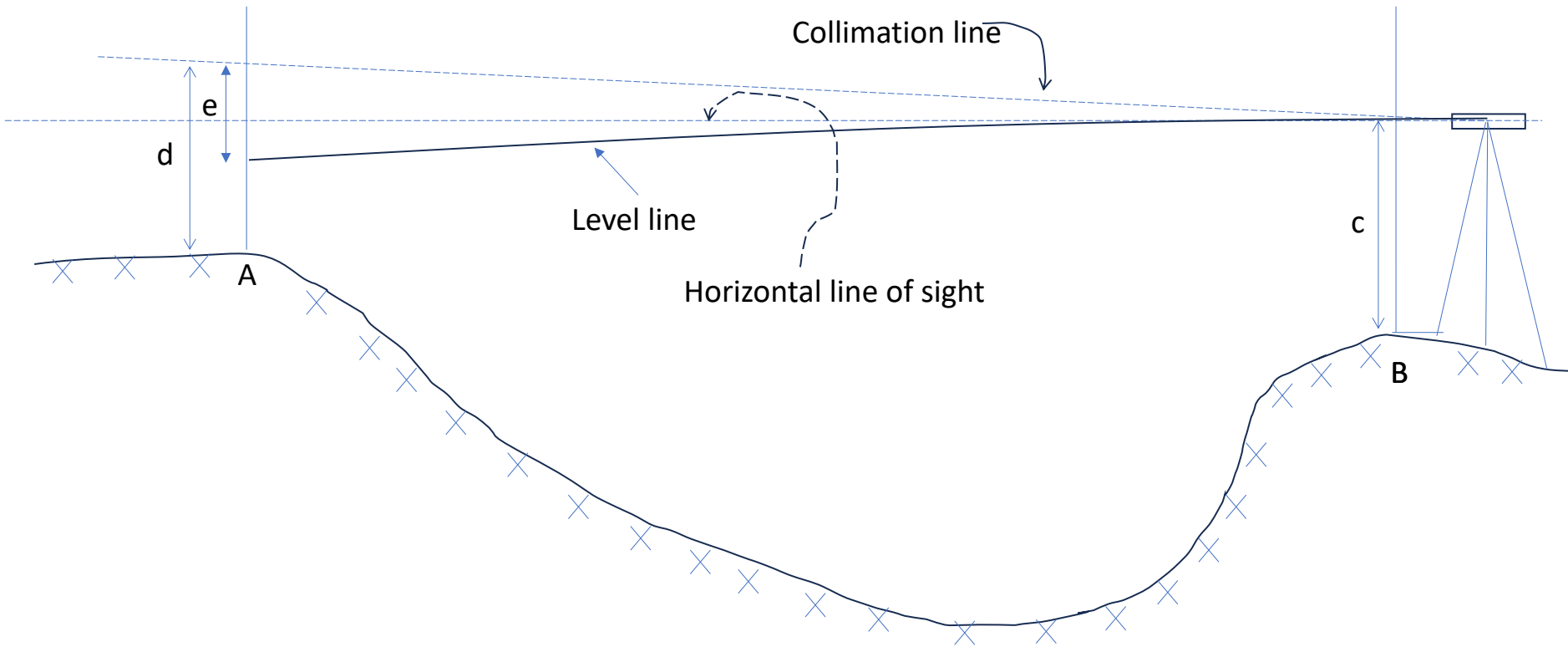
If the instrument axis is not truly horizontal when it is levelled, the collimation error is encountered.

Reciprocal levelling

It is useful where it is not possible to set up the instrument between two stations due to obstruction (like river, valley). Where the sights are also longer, the reciprocal levelling is used.



Instrument is near to Station A.
Collimation line is assumed upward.



Instrument is near to Station B.
Collimation line is assumed upward.

Difference in RL of A and B (h)

$h = \text{RL of A} - \text{RL of B}$

= correct staff reading at B – correct staff reading at A

1st case , $h = (b - e) - a$ - (1)

2nd case, $h = c - (d - e)$ - (2)

$h = \frac{(b-a) + (c-d)}{2}$, eliminating e

$e = \frac{(b-a) - (c-d)}{2}$, eliminating h, error includes error due to curvature, refraction and collimation

Q4: Determine the correction due to Curvature and refraction if the length of sight is (i) 1200 m and (ii) 1800 m.

Solution

(i) $D = 1200 \text{ m} = 1.2 \text{ km}$;

curvature correction = $0.0785 D^2 = 0.0785 \times 1.2^2 = 0.113 \text{ m}$.

refraction correction = $0.0112 \times 1.2^2 = 0.016 \text{ m}$.

ii) $D = 1800 \text{ m} = 1.8 \text{ km}$;

curvature correction = $0.0785 \times 1.8^2 = 0.254 \text{ m}$

refraction correction = $0.0112 \times 1.8^2 = 0.036 \text{ m}$.

Q5: The following observations are made in a reciprocal levelling operation with an auto level and a levelling staff.

Instrument station near	Staff reading on station (in m)		Remark
	X	Y	
X	1.310	2.315	The distance from X and Y is 800.8m.
Y	2.585	1.295	

Find the correct staff reading at X when the instrument is at Y. Find the collimation error.

$$\textcircled{2} \quad a = 1.310 \quad b = 2.315$$

$$d = 2.585 \quad c = 1.295$$

$$\text{Distance between X \& Y} = 800.8\text{m} = 0.8008\text{ km}$$

(i) When instrument is at X

$$h = \text{RL of X} - \text{RL of Y}$$

$$h = (b - e) - a \quad \text{--- (1)}$$

(ii) When instrument is at Y

$$h = \text{RL of X} - \text{RL of Y}$$

$$h = c - (d - e) \quad \text{--- (2)}$$

① + ②

$$2h = (b - e) - a + c - (d - e)$$

$$2h = b - e - a + c - d + e$$

$$h = \frac{(b - a) + (c - d)}{2}$$

$$h = \frac{(2.315 - 1.310) + (1.295 - 2.585)}{2} \quad \textcircled{2}$$

$$h = -0.1425$$

∴ Station Y is @ higher level.

$$\text{Total error } (e) = \frac{(b - a) - (c - d)}{2}$$

$$= \frac{(2.315 - 1.310) - (1.295 - 2.585)}{2}$$

$$e = \frac{1.1425}{2} \quad \textcircled{2}$$

Combined error due to

$$\text{Curvature \& refraction} = 0.0673 D^2 \quad (D \text{ in km})$$

$$= 0.0673 (0.8008)^2$$

$$= 0.0431 \text{ m} \quad \textcircled{2}$$

(i) The correct staff reading = $d - e$

@ X when instrument @ Y

$$= 2.585 - 1.1425$$

$$= 1.4325 \quad \textcircled{2}$$

(ii) Collimation Error

$$\text{From eq (2)} \Rightarrow h = c - (d - e)$$

$$h = c - [d - \{e_{\text{combined}} + e_{\text{collim}}\}]$$

$$-0.1425 = 1.295 - [2.585 - \{0.0431 + e_{\text{collim}}\}]$$

$$e_c = 1.1044 \quad \textcircled{2}$$

(02)

Total Error = Combined error due to Curvature & refraction + Collimation Error

$$e = e_{\text{combined}} + e_c$$

$$e_c = 1.1475 - 0.0431$$

$$= \underline{\underline{1.1044}}$$

Q6: The following observations are made in a reciprocal levelling operation,

Instrument station near	Staff reading on (in m)		Remark
	A	B	
A	1.340	0.895	The distance from A to B is 1255.54m. The R.L. of point B is 355m.
B	1.455	0.755	

Find the following,

- I. The true R.L. of A,
- II. The correction for curvature,
- III. The correction for refraction,
- IV. The collimation error,
- V. State whether the collimation line is bent-up or bent-down.

$$h = R.L. \text{ of } A - R.L. \text{ of } B$$

$$= \frac{(b-a) + (c-d)}{2} = \frac{(0.895 - 1.340) + (0.755 - 1.455)}{2}$$

$$= -0.5725 \text{ m} \rightarrow \text{point B is at higher level than point A}$$

$$\text{So, } -0.5725 = R.L. \text{ of } A - R.L. \text{ of } B$$

$$\Rightarrow R.L. \text{ of } A = R.L. \text{ of } B - 0.5725 = 354.4275 \text{ m}$$

$$\text{(ii) } C^2 \text{ Curvature correction} = 0.0785 \times D^2 \text{ m (-ve)}$$

$$= 0.0785 \times (1.25554)^2 \text{ m}$$

$$= 0.1237 \text{ m (-ve)}$$

$$\text{(iii) Correction for refraction} = 0.0176 \text{ m (+ve)}$$

(iv) Collimation error: (e_c), let's assume collimation line is bent-up. So,

$$h = R.L. \text{ of } A - R.L. \text{ of } B$$

$$= \text{Correct staff reading at B} - \text{correct staff reading at A}$$

$$\Rightarrow -0.5725 = (0.895 - 0.1237 + 0.0176) - e_c$$

$$- 1.340$$

$$\Rightarrow -0.5725 = -0.5511 - e_c$$

$$\Rightarrow e_c = 0.0214 \text{ m}$$

Assumption of bent-up is correct.

(v) Collimation line is bent-up.

innovate

achieve

lead

Contouring

Contour: An imaginary line on the ground surface joining the points of equal elevation (Reduced Level or RL) is known as contour.

In other words, **contour line** is the intersection of a level surface with the surface of earth.

(Contour lines on a plan gives the information about the ground)

Contour Map: A map showing contour lines is known as Contour map.

A contour map gives an idea of the altitudes of the surface features as well as their relative positions in plan, which serves the purpose of both, a plan and a section.

Purpose of counterling

Contour survey is carried out at the starting of any engineering project such as a road, a railway, a canal, a dam, a building etc.

1. To get the topographical information about the ground.
2. To prepare the contour maps in order to select the most economical or suitable site for the projects.
3. To locate the alignment of a canal so that it should follow a ridge line.
4. To mark the alignment of roads and railways so that the quantity of earthwork both in cutting and filling should be minimum.
5. To estimate the reservoir capacity.
6. To trace a route of particular grade.
7. To distribute pipes effectively in water supply projects, collection of sewerage waste.

Others....

Contour Interval: The vertical distance between two consecutive contours is called the contour interval.

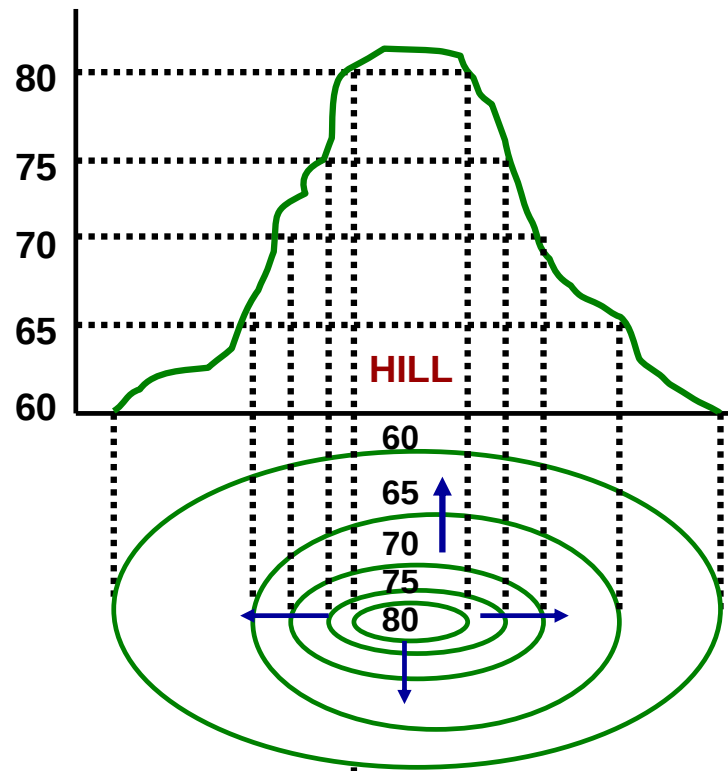
Horizontal Equivalent: The horizontal distance between two consecutive contours is called the horizontal equivalent.

[Normally, the contour interval is constant throughout the map, while the horizontal equivalent is variable and depends upon the stiffness of the ground.]

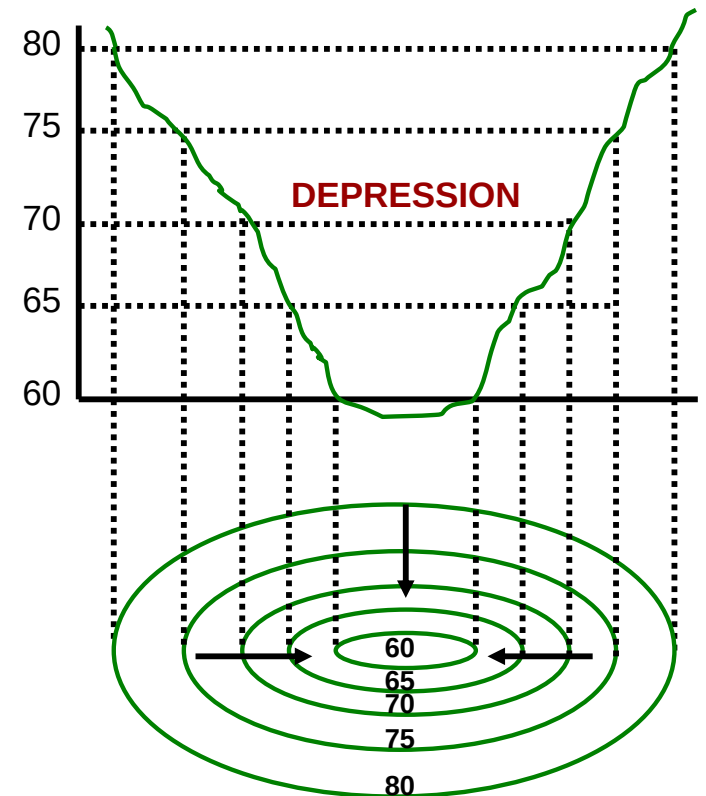
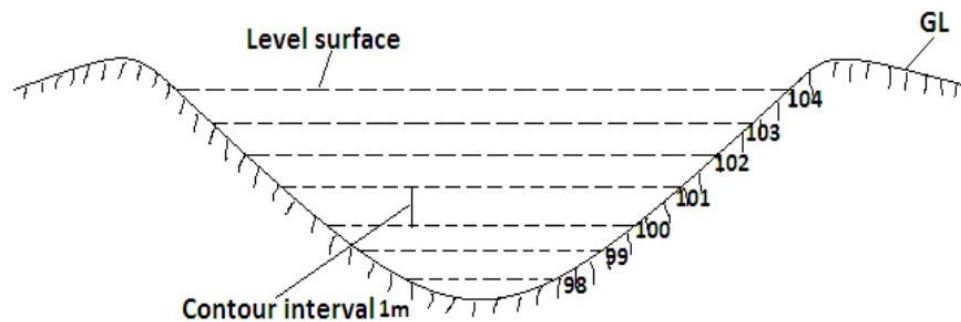
CHARACTERISTICS OF CONTOURS

1. All points in a contour line have the same elevation.
2. In flat ground the contours are widely separated and steep-slope the contours are close to one another.
3. In uniform slope the contour lines are uniformly spaced.
4. In plane surface the contours are straight, parallel and equally spaced.

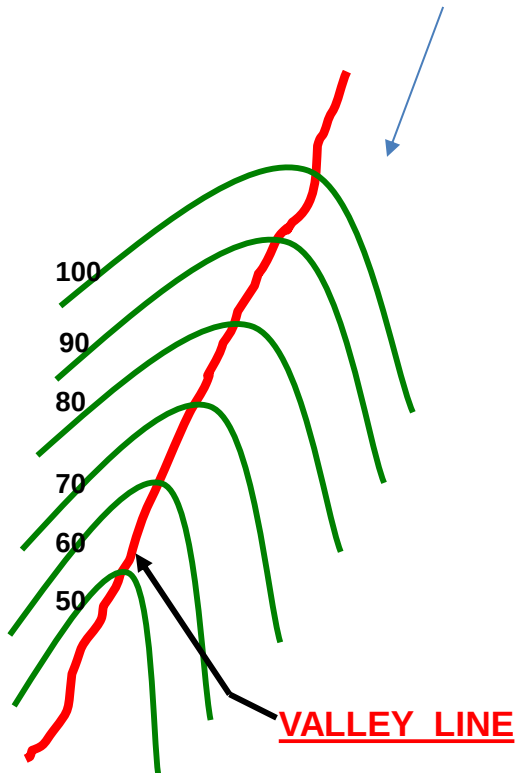
5. A set of closed contour lines on the map represent a hill , if the higher values are inside.



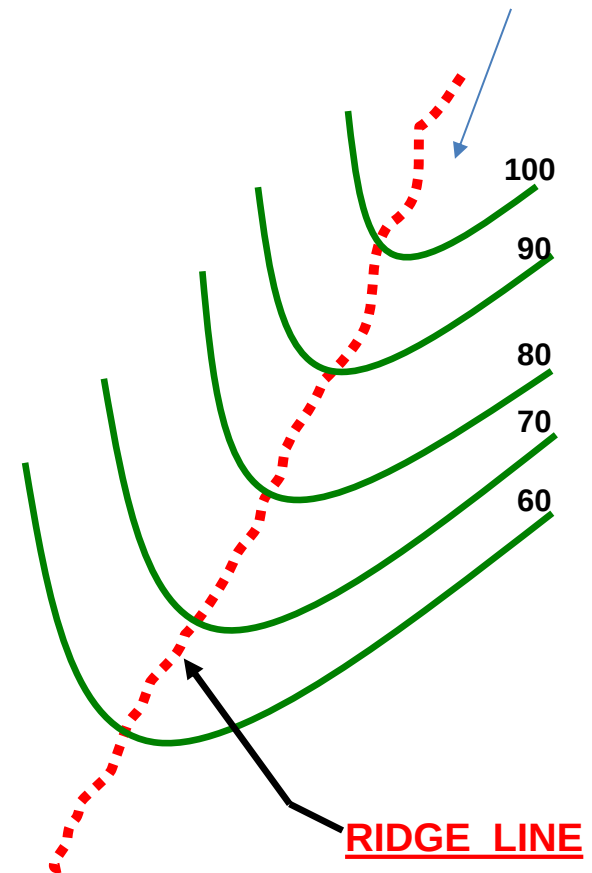
- A set of closed contour lines on the map indicate a depression if the higher values are outside.



7. Contour line cross ridge or valley line at right angles

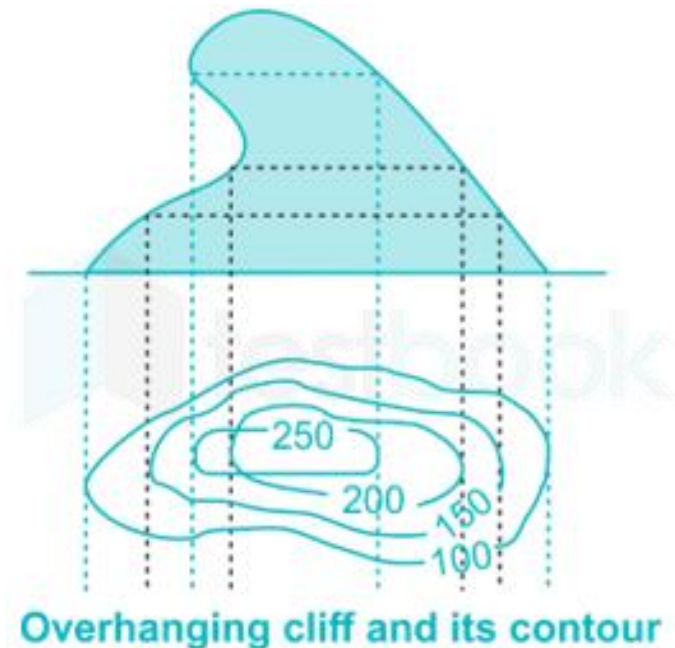
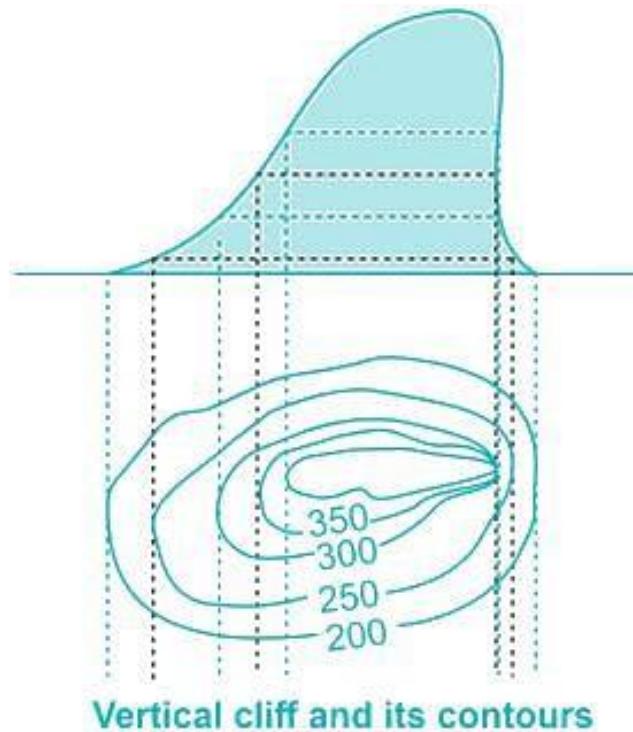


Higher values are outside the bend.

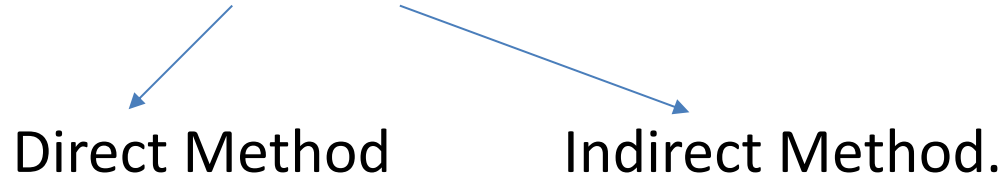


Lower values are outside the bend.

8. Contours do not end anywhere but close on themselves either within or outside the limits of the map.
9. Contour lines do not merge or cross one another on map except in the case of an overhanging cliff and vertical cliff.



Methods of contouring

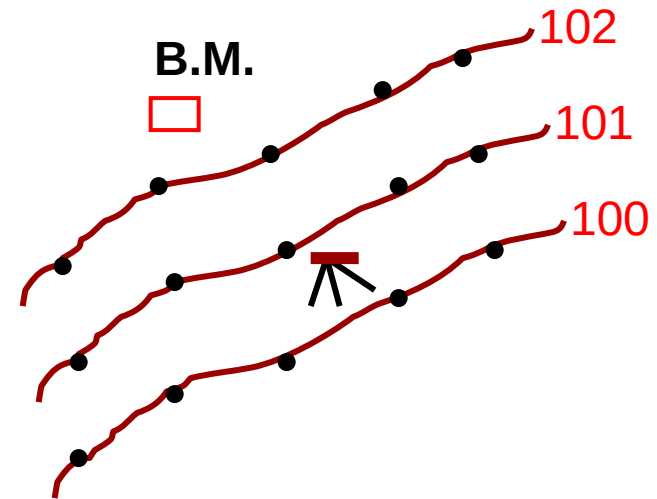


Direct Method

Indirect Method.

Direct Method

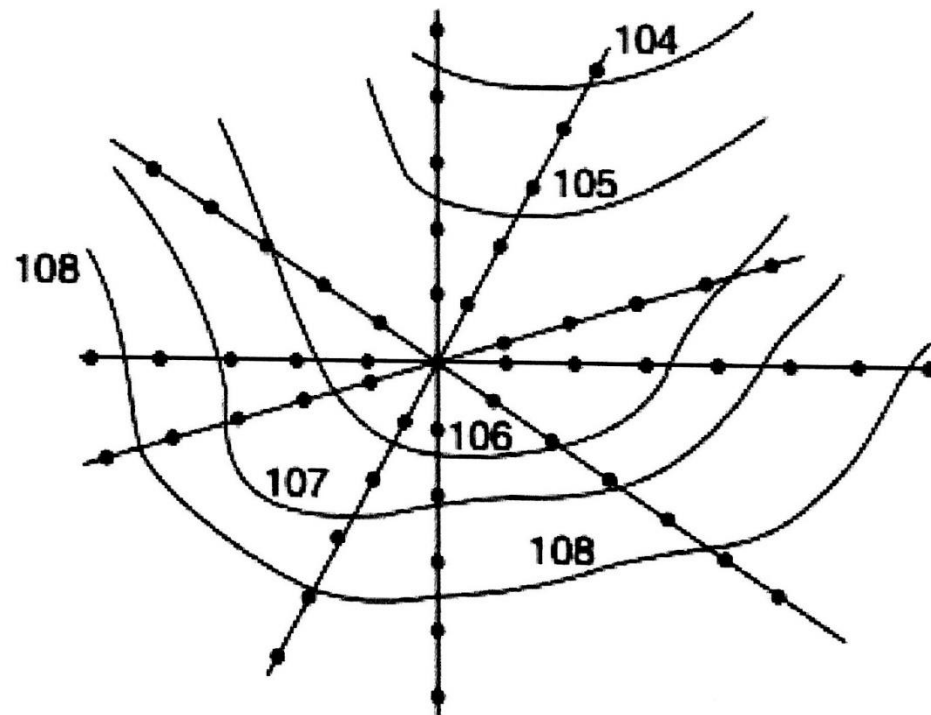
A method that uses field measurements to plot contours on the ground itself.



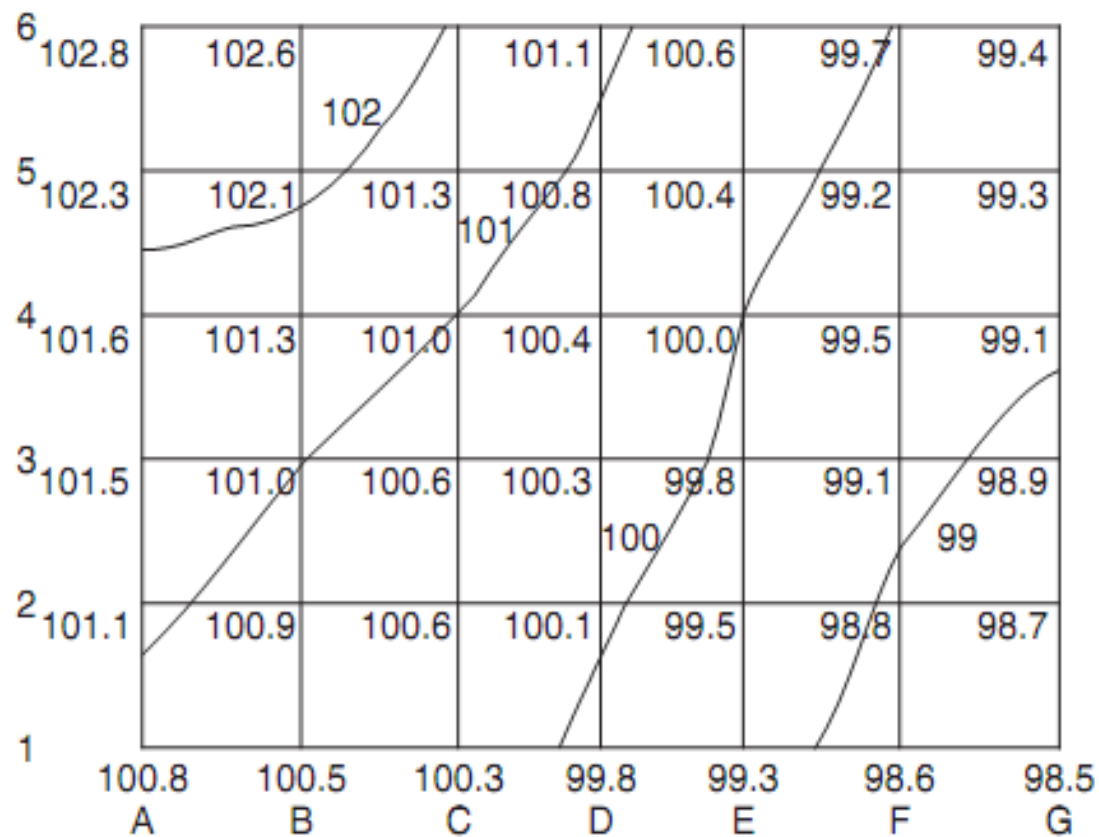
By finding the points of equal elevation by hit and trial method and marking those. This is done by finding points with equal staff reading.

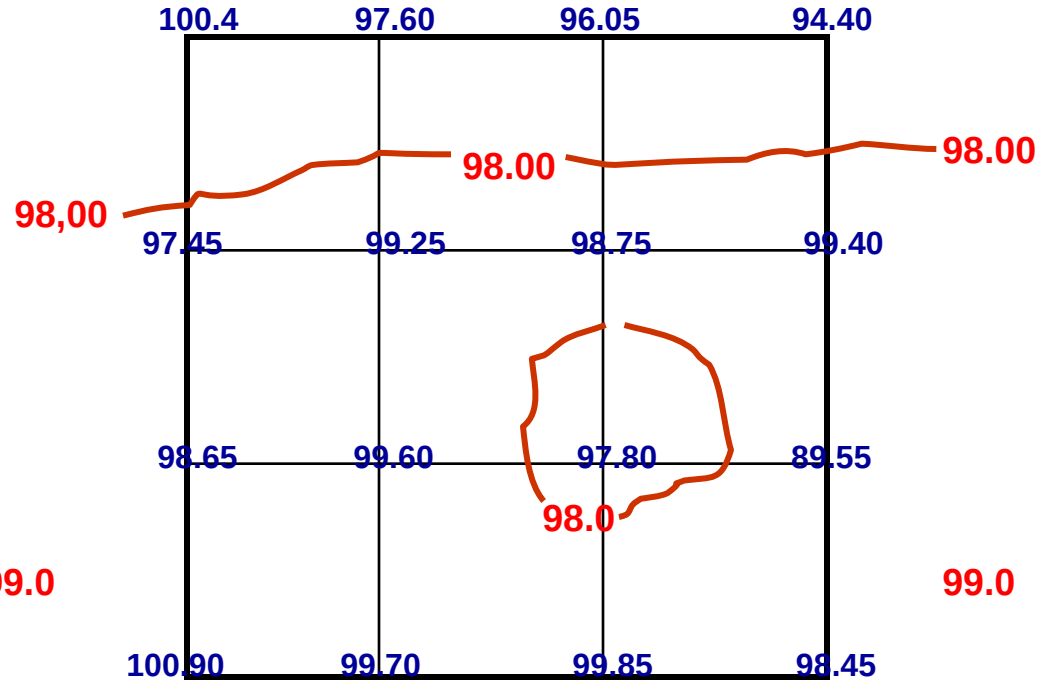
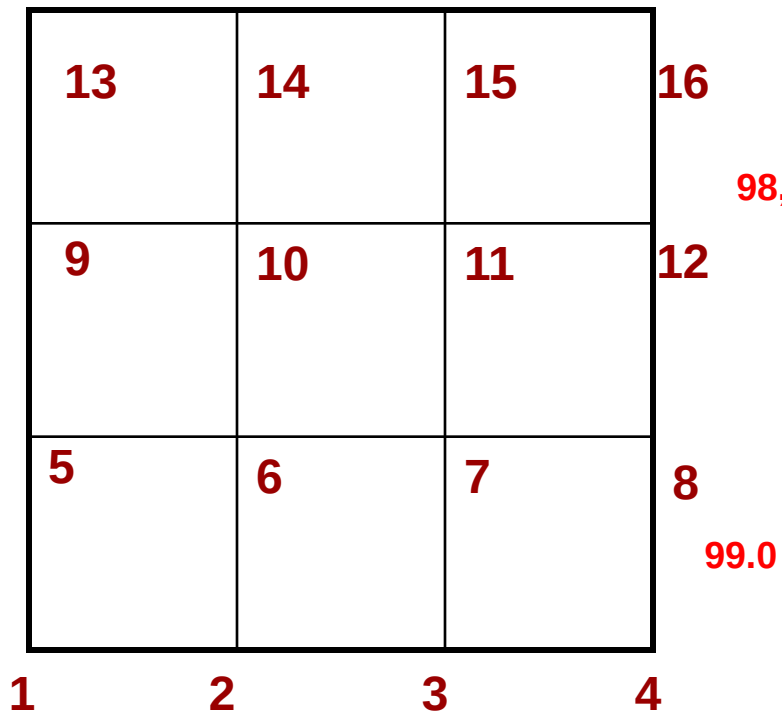
- This method is most accurate but very slow and tedious as *a lot of time is wasted* in searching points of the same elevation for a contour.
- This is suitable for small area and where better accuracy is required.

Direct Method(using radial lines)

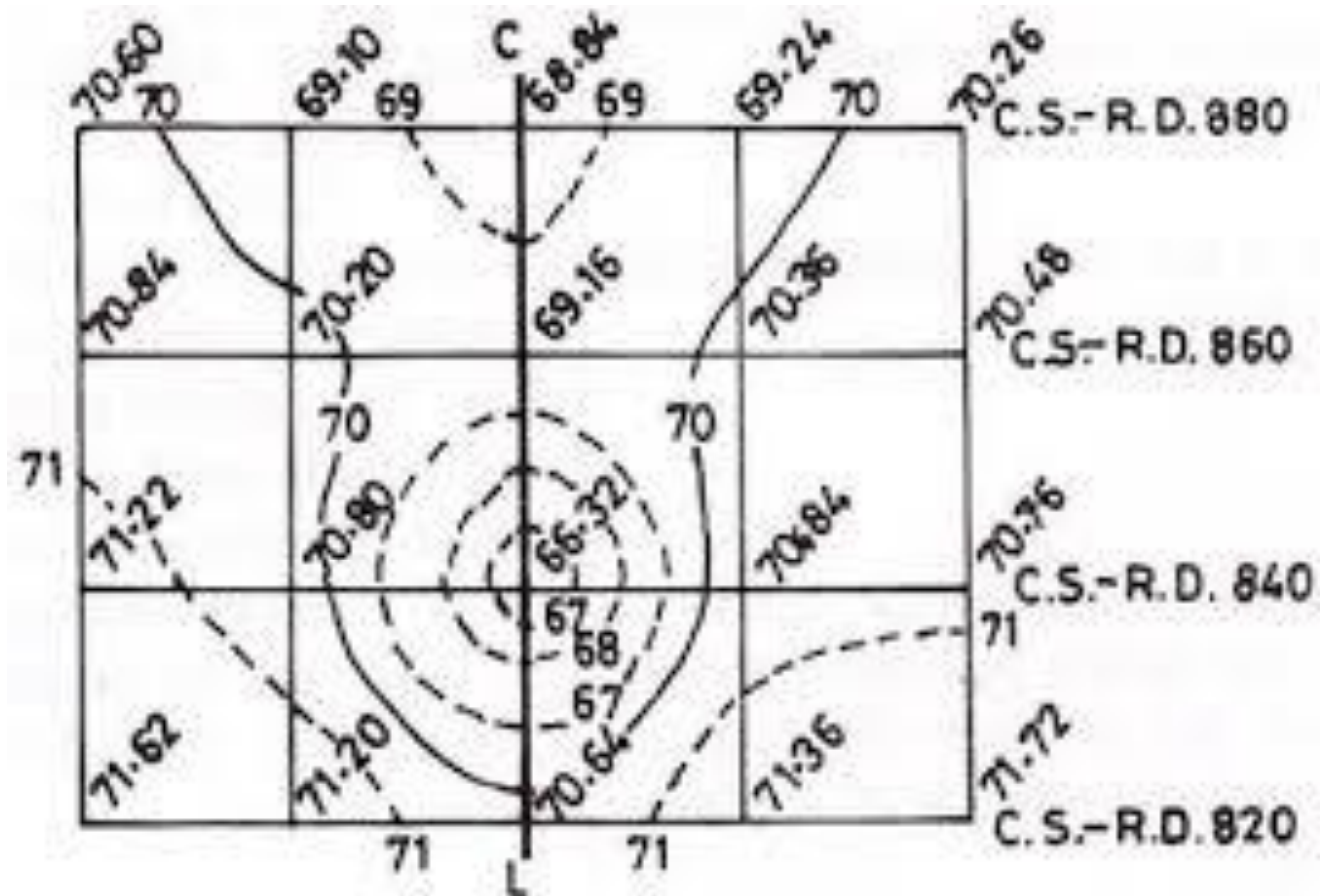


Indirect Method (method of square)





Indirect Method (cross-section method)

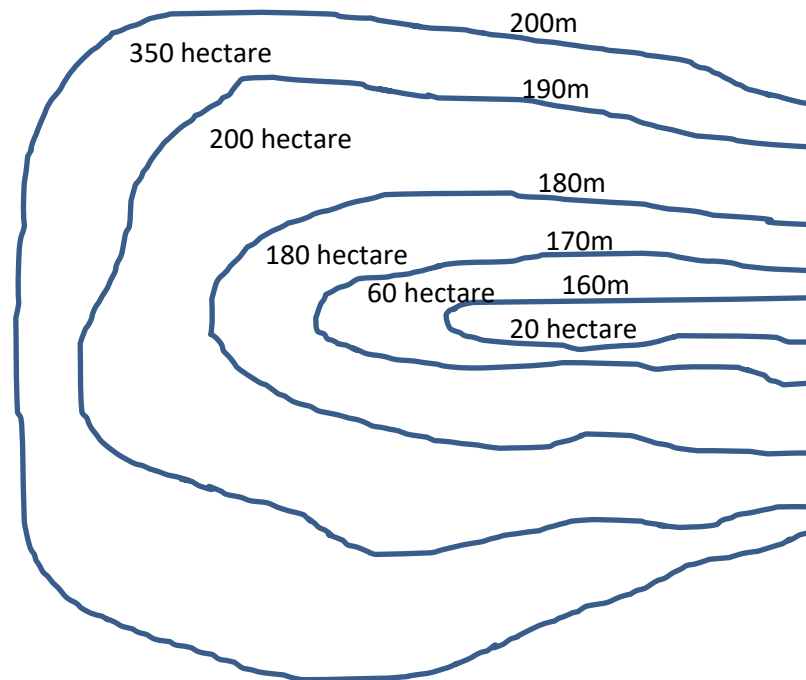


Q1.

100.2	100.350	99.235	98.755	100.345	101.005	100.755	102.355	99.995
99.850	100.540	100.670	99.210	98.750	99.230	100.570	101.700	
100.90	101.545	99.690	103.220	98.755	100.250	101.990	98.900	
99.830	101.700	98.980	101.120	99.890	99.290	100.980	99.780	
102.00	101.00	100.760	101.560	99.780	101.380	100.155	99.090	
101.100	100.990	100.790	102.780	99.670	101.400	100.655	100.900	
102.340	98.670	102.895	101.2300	98.765	101.605	100.765	99.800	
101.40	98.775	99.795	101.245	98.770	102.355	101.875	101.955	
100.340	101.980	99.665	101.540	98.980	100.550	101.115	100.855	
98.70	100.795	101.795	101.765	99.250	100.675	101.255	101.545	

Consider the length of the stations on both directions as 5m. Plot the contours of 99.5, 100, 100.5, 101.0.

Q2. In the contour map of the reservoir the RL of the different contour lines and the area contained by these lines are given. Find the volume of the water of the reservoir between the contour line with RL=200m and the contour line with RL=160m.



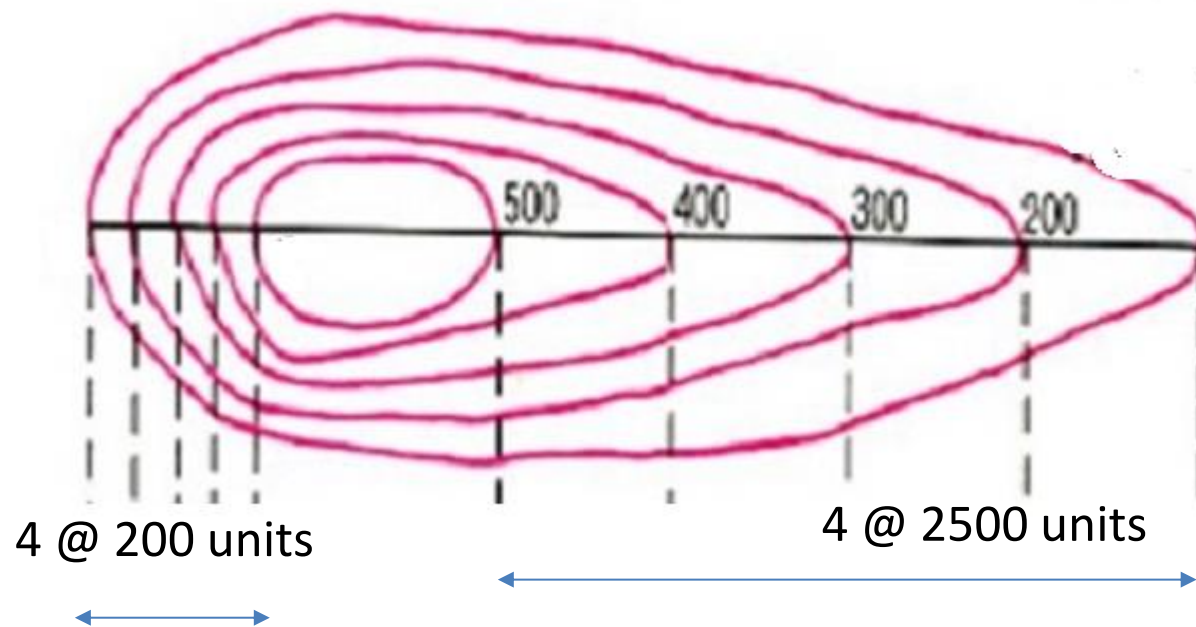
Solution:

Contour interval = $d = 10\text{m}$

$A_1 = 350$ hectare, $A_2 = 200$ hectare, $A_3 = 180$ hectare, $A_4 = 60$ hectare, $A_5 = 20$ hectare

Volume by trapezoidal rule = $d \cdot \frac{A_1 + A_2}{2} + d \cdot \frac{A_2 + A_3}{2} + d \cdot \frac{A_3 + A_4}{2} + d \cdot \frac{A_4 + A_5}{2} = 6250 \text{ hectare.m}$

Q3.



Find the slope of the ground in left side and right side. Both slopes are uniform.

Solution

$$\text{Right Side: } \frac{300}{7500} = \frac{1}{25}, \quad 1 \text{ in } 25$$

$$\text{Left Side: } \frac{300}{600} = \frac{1}{2}, \quad 1 \text{ in } 2$$



Thank You